Biochemistry Cumulative Examination

February 28, 1998

1. One common method for activation of a particular enzyme is the "cascade." In a cascade enzyme A activates enzyme B which in turn activates enzyme C which then acts on another enzyme and so on. Only after going through multiple steps is the enzyme that actually does something of importance activated. This seems inefficient. What advantages do cascades offer over simply directly activating an enzymatic activity? (15 points)

2. Name an example of an activation cascade. (10 points)

3. Name two different mechanisms by which one enzyme can activate another. (10 points)

4. Suppose you discover a new metabolic pathway. The starting compound in this new pathway, Stuff, is a well known metabolic intermediate in another pathway. As the first step in the new pathway you have found, Stuff is converted into Junk in a very favourable reaction (-10 kcal/mol). The enzyme Stuffase carries out this reaction. Junk is then turned into iso-Junk by Junk Mutase, a reaction that is near equilbrium (0 kcal/mol). In the last step of this pathway iso-Junk is transformed into Product, coupled to the hydrolysis of ATP to ADP and P_i, in a mildly favourable overall reaction (-2 kcal/mol) catalyzed by Product Synthetase. As far as you can tell, Junk and iso-Junk have no other use in the cell. Given this information, if this pathway were regulated at only one point, which enzyme would you expect to be regulated? Explain your reasoning. (25 points)

5. Some enzymes seem to be self-regulating. When the level of substrate is low the ezymatic activity of the enzyme is low. When the substrate level builds up past some critical level, the enzyme becomes much more active. What causes this sort of effect and what is the name for it? (20 points)

6. Suppose that the activity of an enzyme you are studying in *E. coli* is virtually absent under normal growth conditions. When you add the substrate to the growth you see that enzyme activity appears in the bacteria, but only 10 or 20 minutes after you add substrate. Activity continues to climb for another 2 or 3 hours and then plateaus. If you remove substrate the level of enzyme activity drops over the next 4 or 5 hours until it finally disappears. Explain what is likely occuring. (20 points)